Multiple Player Tracking in Sports Video: A Dual-Mode Two-Way Bayesian Inference Approach With Progressive Observation Modeling

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***Abstract*—Multiple object tracking (MOT) is a very challenging task yet of fundamental importance for many practical applications. In this paper, we focus on the problem of tracking multiple players in sports video which is even more difficult due to the abrupt movements of players and their complex interactions. To handle the difficulties in this problem, we present a new MOT algorithm which contributes both in the observation modeling level and in the tracking strategy level. For the observation modeling, we develop a progressive observation modeling process that is able to provide strong tracking observations and greatly facilitate the tracking task. For the tracking strategy, we propose a dual-mode two-way Bayesian inference approach which dynamically switches between an offline general model and an online dedicated model to deal with single isolated object tracking and multiple occluded object tracking integrally by forward filtering and backward smoothing. Extensive experiments on different kinds of sports videos, including football, basketball, as well as hockey, demonstrate the effectiveness and efficiency of the proposed method.**

***Index Terms*—Bayesian inference, object detection, object tracking, observation modeling, sports video.**

I. Introduction

With the explosive growth of various video data, automatic video content analysis becomes an extremely urgent task. Among all kinds of video data, sports videos captured from different kinds of matches, such as football and basketball, attract much more attention. This is due to not only their great popularity, but also the tremendous commercial value. Automatic analysis of sports video data, such as highlights extraction, technical statistics gathering, and tactical analysis, can provide great convenience for both video content service providers and sports game watchers. To accomplish these tasks automatically, locating and tracking multiple players within the playfield region plays a very important role which can provide many fundamental results for them.

In this work, we aim to track multiple highly dynamic and highly interactive players in sports videos. Compared with multiple object tracking (MOT) in other scenes, multiple player tracking in sports video is much more difficult due to the following reasons: 1) abrupt changes in view and pose of one single player; 2) complex interactions and severe occlusions among multiple players; 3) and various influences from other factors, such as disturbances of nonplayer objects, large camera zooming changes, and motion blur. All of these issues together have posed quite a great challenge to the tracking system, which should possess not only remarkable capability for observation collection but also a sophisticated tracking strategy to make the system robust under different situations. In this paper, we innovate from both of the observation modeling level and the tracking strategy level to build a robust multiple player tracking system on different types of sports videos. The main contributions of this paper include those given here.

- A new progressive observation modeling process to distinguish a target from other objects step by step. This progressive mechanism not only transforms a complicated problem to several solvable subproblems, but also provides many important intermediate results for tracking.
- A novel dual-mode two-way Bayesian inference framework for the tracking problem. This framework copes with the single isolated object tracking and multiple occluded object tracking problems in a unified manner and also combines the advantages of the forward filtering and backward smoothing at the same time.
- A robust and extendible MOT system which can successfully track arbitrary number of players on many typical kinds of sports videos, like football, basketball and hockey ball. Compared with other state-of-the-art methods on player tracking, the proposed system has achieved superior tracking performance with very satisfactory processing speed.

The remainder of this paper is organized as follows. Related work is discussed in Section II. Problem formulation is derived in Section III. A detailed description of the approach is presented in Section IV, where the progressive observation modeling process is developed in Section IV-A, and the tracking strategy using dual-mode two-way Bayesian inference is described in Section IV-B. Experimental results and analysis are given in Section V, and Section VI concludes the whole paper.

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II. Related Work

In object tracking, observation model and tracking strategy are two key components. For multiple player tracking in sports video, even more effort is required to build the observation model and to design the tracking strategy. Here, we first make a short summation of different kinds observation models used in previous works. We then we give a concise review of some representative tracking strategies. Finally, we discuss some previous works which are also targeted at player detection and tracking in sports video.

A. Observation Model

With the fast development of object tracking research, various kinds of observation models have been used for object tracking in previous work, such as feature points [1]–[3], templates [4], and moving areas [5]. Recently, the fast development of object detection techniques has resulted in many promising methods for detecting particular object classes, e.g., faces [6], [7] and pedestrians [8], [9]. These object detectors provide good observation models for detection-based tracking algorithms. By making the detectors output confidences for soft decision, they can be used for object trajectory initialization, growth, and termination [8], [10]. However, the detectors are far from perfect. Miss detections, false alarms, and inaccurate detection responses are three main problems that bother many detection based object tracking algorithms. To enhance the observation model, many methods fuse multiple cues concurrently [4], [8], [11], such as object appearance, shape, and motion. Evaluating the effectiveness of various cues and how to organize them, however, still require further study.

B. Tracking Strategy

Once having object observations, the next problem is how to use them in a tracking system. Generally, there are two different strategies. One is to use them from a probabilistic inference perspective. A Kalman filter [12] is an early representative method of this strategy. Recently, based on sequential Monte Carlo sampling, particle filters [13] gained much attention because of their simplicity, generality, and extensibility in a wide range of challenging applications. One successful application of particle filter is for single object tracking (SOT). In [14], Shen *et al.* fused multiple cues in a particle filter framework to track a single object. Li *et al.* [10] used multiple detectors to form a cascade particle filter to track one face. The order of applying the detectors was determined based on their computational costs and discrimination power. Some work tried to extend the particle filter to MOT. For example, Hue *et al.* [15] used a stochastic vector to assign the observation in the particle filter process. Xing *et al.* [11] improved the particle filter to select multiple part detectors to track multiple objects with different degrees of occlusion. To address the problem of multiple modes of a target, Vermaak *et al.* [16] used a joint representation of the target state space. However, using a unified observation model for all of the objects still cannot distinguish them well, especially when occlusions are presented. To further cope with the multiple object occlusion problem, global association-based tracking methods provide another new strategy from the deterministic optimization perspective. Given the detection responses, these methods try to link them globally to handle the multiple objects interaction problem by tailoring a certain kind of optimization algorithm, such as the Viterbi algorithm [17], linear programming [18], min-cost flow [19], and the Hungary algorithm [20]. Since the information of all of the frames are available before tracking, these methods could reach global optimum theoretically. The main drawback, however, is the heavy computation and the temporal delay that limit their use for real-time applications.

C. Player Detection and Tracking in Sports Video

There have been only a handful of papers concerned with multiple player detection and tracking in sports video, among which football players and hockey players are two main explored targets. To detect the players, one method is to use color and shape information. In [21], Huang *et al.* first detected the players and the ball based on extracted foreground and then performed shape analysis to remove false alarms. Pallavi *et al.* [22] used a directed weighted graph to detect and track multiple players based on grouped pixels from a region growing algorithm. This kind of method obtained reasonable results for some sports videos. However, color and shape information only provide a descriptive observation of the players and are sensitive to noise. To address these limitations, learning-based player detection provides a more discriminative observation over this problem and is much more popular [23]–[28]. In [26], Zhu *et al.* used a support vector machine (SVM)-based detector to detect football players and initialized a support vector regression (SVR) particle filter for each player to be tracked, while, in [23], Breitenstein *et al.* employed a boosting method to detect players and performed online single-frame association to track them. To surmount the problem of complex multiple object interactions with overlaps and ambiguities, Okuma *et al.* [25] used an offline boosted particle filter to track each player in a mixture representation as in [16]. An offline boosted particle filter can be adapted to various changes of players but may be less effective at distinguishing different players. Tracking multiple highly dynamic and interactive players in sports videos is still a very challenging problem that is far from being solved.

III. Problem Formulation

Here, we will derive our unified Bayesian formulation of the SOT and MOT problems. Meanwhile, some unique properties of this unified formulation will also be discussed.

A. Object Tracking as Sequential Bayesian Estimation

Denoting the object state sequence as $\mathbf{s}_{1:t} = \{\mathbf{s}_1, \ldots, \mathbf{s}_t\}$, which is usually assumed to be a Markovian process and the observation sequence as $\mathbf{o}_{1:t} = \{\mathbf{o}_1, \ldots, \mathbf{o}_t\}$, object tracking is usually formalized as a sequential Bayesian estimation problem by a two-step recursion of Prediction (P) and Update (U) as follows:

$$P : p(\mathbf{s}_t|\mathbf{o}_{1:t-1}) = \int D(\mathbf{s}_t|\mathbf{s}_{t-1})p(\mathbf{s}_{t-1}|\mathbf{o}_{1:t-1})d\mathbf{s}_{t-1} \quad (1)$$

$$U : p(\mathbf{s}_t|\mathbf{o}_{1:t}) \propto L(\mathbf{o}_t|\mathbf{s}_t)p(\mathbf{s}_t|\mathbf{o}_{1:t-1}) \quad (2)$$

where $D(\mathbf{s}_t|\mathbf{s}_{t-1})$ is the dynamic model of the system that reflects the tracking strategy and $L(\mathbf{o}_t|\mathbf{s}_t)$ is the observation model that gives the likelihood of observation in the state space. Here, we can further understand the importance of the observation model and tracking strategy to a tracking system. If we want to improve the performance of a tracking system, both of these two components need to be elaborately designed.

In implementation, the filtering distribution $p(\mathbf{s}_t|\mathbf{o}_{1:t})$ can be approximated by a set of weighted particles $\{\mathbf{s}_t^n, \pi_t^n\}$ in the sequential Monte Carlo framework (also known as particle filter) [13]

$$p(\mathbf{s}_t|\mathbf{o}_{1:t}) = \sum_{n=1}^{N} \pi_t^n \delta_{\mathbf{s}_t^n}(\mathbf{s}_t) \tag{3}$$

where N is the total number of particles and $\delta_{\mathbf{s}}(\cdot)$ denotes the delta–Dirac function at position $\mathbf{s}$.

This formulation is popular for SOT and has showed great successes in tracking objects like hand [13], face [10], and head [4]. When applying it to MOT, however, it has at least two problems. One is how to assign the corresponding observations to each object when multiple objects interact with each other. We refer to it as the *observation assignment* problem, which usually causes an ID switch when tracking two adjacent objects. The other is how to proceed with tracking when no observations could be collected for one object because of occlusion or abrupt motions. We refer to it as the *observation missing* problem, which usually causes a fragmentation on one object trajectory. These two problems are much more prevalent in multiple player tracking due to their highly dynamic and highly interactive movements. We will develop our formulation to address these two problems in Section IV.

B. Object Tracking as Dual-Mode Two-Way Bayesian Inference

To extend the above model to MOT, there are usually two formulations. One is to initialize a new track for each object and track them independently [10], [25], [26]. The other is to view all of the objects as a whole and track them globally [19], [20]. On the one hand, while the former formulation is easy to understand and implement, it is still very difficult to deal with the observation assignment problem. On the other hand, the latter formulation can give a more general representation, but it often makes the problem intractable or requires heavy computation. In this paper, we follow a divide-and-conquer strategy to formulate the MOT problem as a union of single isolated object tracking and multiple occluded object tracking. Denoting $\mathbf{S}_t = \{\mathbf{s}_{t,m}\}_{m=1}^{M}$ as the state set of all of the M objects at time t, we divide the object set into two subsets according to their spatial relations: the isolated object set $\mathbf{S}_t^-$ and the occluded object set $\mathbf{S}_t^+$. For each object in $\mathbf{S}_t^-$, we track it independently with a general observation model which is designed to distinguish them from the backgrounds. For the objects in $\mathbf{S}_t^+$, we view each occlusion cluster as a whole and track them simultaneously with a dedicated observation model which is designed to distinguish different players. Suppose there are M_z occluded objects in one occluded cluster C_z at time t, $\mathbf{s}_{t,C_z}$ is the object states, and $\mathbf{o}_{t,C_z}$ is the supporting observations. The corresponding filter distribution can be approximated as

$$p(\mathbf{s}_{t,C_z}|\mathbf{o}_{1:t,C_z}) = \sum_{n \in C_z} \pi_t^n \delta_{\mathbf{s}_{t,C_z}^n}(\mathbf{s}_{t,C_z}). \tag{4}$$

Now, the observation assignment problem can be naturally handled by making the candidate model collect supporting observations for each object. Next, we will address the observation missing problem which is done by introducing the smoothing concept into the tracking problem.

The above mentioned formulation of the tracking problem is also known as forward filtering in the estimation problem. This corresponds to estimating the distribution of the current system state based upon observations received up until the current time. This formulation makes the tracking system adapt quickly to changes and easy to implement. The main problem, however, is its over-sensitivity to temporal observation missing. If one object cannot receive observations in several frames, the forward filtering process will lose it even if the object can be reobserved later. In this work, we surmount this problem by introducing a Backward Smoothing procedure to the tracking problem. The basic idea of the backward smoothing procedure is simple: if the system cannot obtain enough observations for one object at current frame, it will wait and re-track the object when enough observations are obtained at a later time. The backward smoothing is another important concept in the estimation problem which corresponds to estimating the distribution of the state at a particular time given all of the observations up to a later time T. For one object $\mathbf{s}_t$, this is tantamount to estimating the distribution $p(\mathbf{s}_t|\mathbf{o}_{1:T})$. Theoretically, this marginal distribution could be obtained directly by integrating out the variables $(\mathbf{s}_{1:t-1}, \mathbf{s}_{t+1:T})$ from the joint distribution $p(\mathbf{s}_{1:T}|\mathbf{o}_{1:T})$ which can be represented as

$$p(\mathbf{s}_t|\mathbf{o}_{1:t}) = \int p(\mathbf{s}_{1:T}|\mathbf{o}_{1:T}) d\mathbf{s}_{1:t-1} d\mathbf{s}_{t+1:T}. \tag{5}$$

Several strategies can be used to estimate this distribution [29] if there is only one object. In our case, however, distributions of multiple unobserved objects need to be estimated simultaneously. Denoting the buffered unobserved object set in the forward filtering procedure as $\mathbf{s}_{t,B}$, the backward smoothing process can be formulated as

$$p(\mathbf{s}_{t,B}|\mathbf{o}_{1:T}) = \frac{p(\mathbf{s}_{t,B}|\mathbf{o}_{1:t-1}) p(\mathbf{o}_{t:T}|\mathbf{s}_{t,B})}{p(\mathbf{o}_{t:T}|\mathbf{o}_{1:t-1})}. \tag{6}$$

Instead of simulating this distribution with particle sampling, we directly find its maximum *a posteriori* probability through a matching process between the distribution $p(\mathbf{s}_{t,B}|\mathbf{o}_{1:t-1})$ and $p(\mathbf{o}_{t:T}|\mathbf{s}_{t,B})$. Based on the matching result, trajectories of the buffered objects that have been successfully retracked are linked and smoothed using the corresponding observation.

By now, we have elaborated our problem formulation. Since it gives a unified Bayesian formulation of the SOT and MOT problems through forward filtering and backward smoothing, we term it as the dual-mode two-way Bayesian inference.

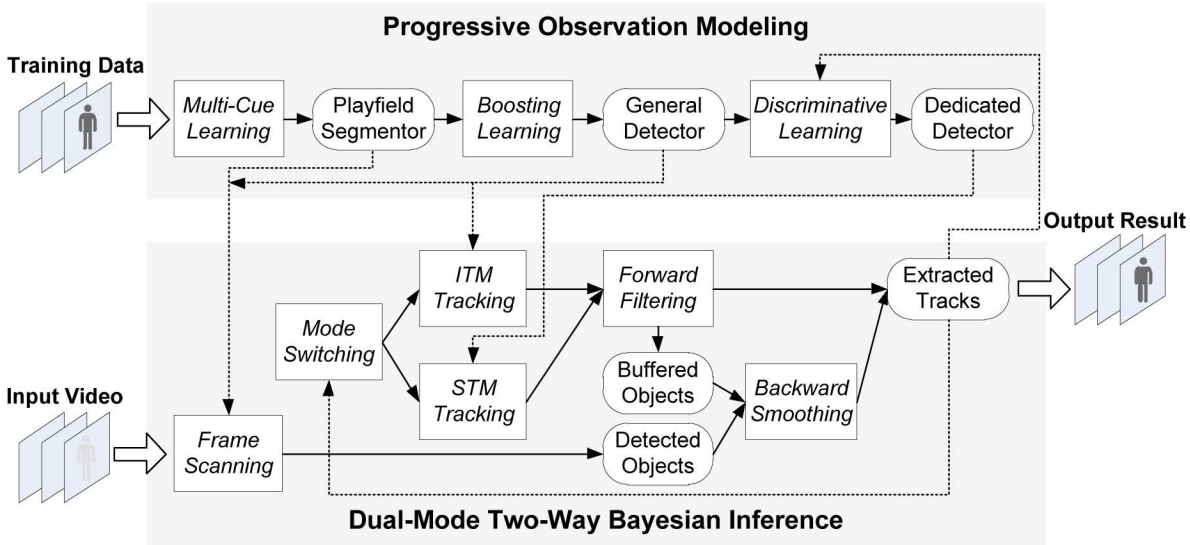

Fig. 1. System flowchart. Rectangle box: procedure; round rectangle box: obtained model; solid arrow: data flow; dotted arrow: model action line; double-line arrow: input and output.

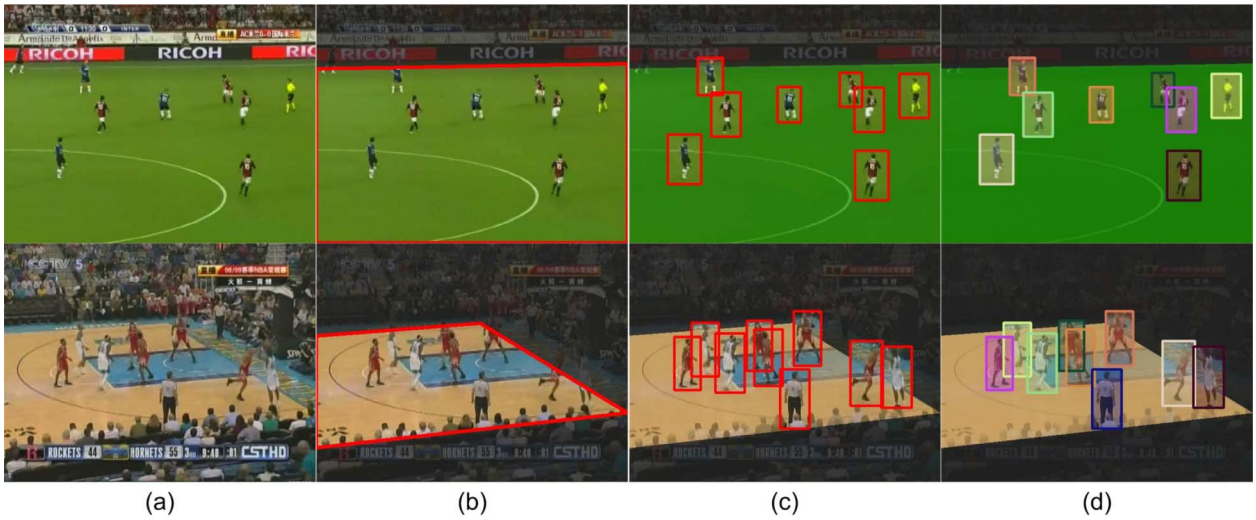

Fig. 2. Progressive observation modeling process. (a) Input frame. (b) Classify playfield and nonplayfield. (c) Classify players and playfield. (d) Classify different players. Each step concentrates on its own task and contributes to the task of the next step. The whole procedure provides many intermediate results that will greatly facilitate the tracking algorithm.

IV. Approach

The proposed MOT system in sports video consists of two main components.

- **Progressive Observation Modeling**. This component builds a strong observation model to provide measurements for the MOT system. In consideration of the complexities of sports videos, three different observation models are built, and each of them provides special functionalities in the system.
- **Dual-Mode Two-Way Bayesian Inference**. Based on the observation modeling component, this component is designed to successfully track different players in the video. This component makes the full use of the results provided by the progressive observation modeling component which is achieved by a Bayesian inference approach that deals with different situations in the tracking scenario.

The entire procedure of the proposed system is summarized in Fig. 1. The details of each component will be discussed below.

A. Progressive Observation Modeling

The key targets of interest in sports videos are the players within the playfield region which reveal the most important part of a game and catch most eyes of watchers. These players, however, are not so easy to be "caught" by a tracking algorithm. To obtain strong observations of the multiple players for tracking, instead of pursuing to directly distinguish players in the playfield from all other objects, we try to distinguish them from other objects step by step. Specifically, in the first step, our method builds an observation model to classify the playfield region and nonplayfield regions, then it focuses on classifying players and the playfield, and at last it aims at distinguishing different players. Fig. 2 gives an illustration of this process. Within

each step of the progressive observation modeling process, different models are designed to fulfill their special tasks.

1) Classifying Playfield and Nonplayfield: The first step of the progressive observation modeling is to distinguish the playfield region from nonplayfield regions. To segment the playfield region from an image, many previous methods use color information to detect the playfield [30]–[33]. The segmentation algorithm is mainly based on dominant color learning [30], [32] or Gaussian mixture models (GMMs) [31], [33]. The objective of our playfield segmentation is not only to locate the dominant playfield region, but also to extract the exact playfield contour. Thus, merely using color is obviously not sufficient.

Here, we design a multicue learning algorithm to obtain a robust playfield segmentor. Our method consists of two stages: the offline stage and the online stage. Within these two stages, multiple useful cues, including color, motion, and shape, are explored and learned to form the final playfield segmentor.

In the offline stage, a spatial temporal GMM (ST-GMM) is learned to classify the playfield. The ST-GMMs collect learning samples for each location in the spatial and temporal domains. Fig. 3 gives an illustration of this sample-collecting strategy. The ST-GMMs can be learned from much less training data without losing the generalization ability and are more robust to noise. Based on the color distributions of collected samples, the ST-GMMs dynamically decide the component number of GMMs. Specifically, during the training process, if the mean values of two Gaussian components are too close (the difference between them is below a threshold T_G), they are merged as one; and when one Gaussian component has a relatively large variance (decided by a threshold T_P), it is split into two. This process iterates until convergence to form the final model. Denoting the learned background and foreground GMMs at location (u, v) as $G_B(u, v)$ and $G_F(u, v)$, respectively, the segmentation result after the ST-GMMs is measured by

$$p_t^c(u, v) = \frac{G_F(u, v; \mathbf{x})}{G_F(u, v; \mathbf{x}) + G_B(u, v; \mathbf{x})} \tag{7}$$

where $G(u, v; \mathbf{x})$ calculates the mixture probability given pixel value $\mathbf{x}$ using the corresponding model.

In the online stage, a motion model is employed to capture the correlation between successive frames. Furthermore, a shape model is adopted to extract the final playfield contour. With the fine segmentation result of previous frame, the motion model first propagates it to the current frame and then uses the propagated result to update the classification result of the offline model. This process can be represented as

$$p_t^m(u, v) = \begin{cases} p_t^c(u, v), & t = 0 \\ (1 - \lambda) p_{t-1}^s(u, v) + \lambda p_t^c(u, v), & t > 0 \end{cases} \tag{8}$$

where $p_t^m(u, v)$ and $p_t^s(u, v)$ are the segmentation results after the motion model and shape model, respectively, and λ is a learning rate. For most sports videos, the playfield region is a rectangle. However, the exhibited shape in the frame is a convex polygon due to the camera view angle and view range. See Fig. 4 for an example. Thus, the shape model models the playfield region as a convex polygon and aims to extract its contour. To do this, it first finds the maximal contour of the updated segmentation result, and then it compacts the extracted contour using the Douglas–Peucker algorithm [34] to remove abrupt edge changes caused by noises. Finally, it calculates the convex hull of the compacted contour as the final playfield shape. Fig. 5 gives an illustration of the whole playfield contour extraction process on two typical kinds of sports video frames.

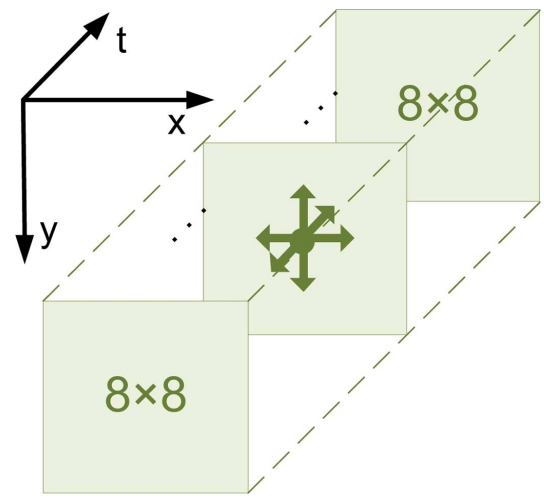

Fig. 3. Illustration of the ST-GMM. To train the GMMs for one location, it collected samples from its neighboring patch in spatial space (e.g., 8×8 patch) and temporal space (corresponding patch location in each frame).

2) Classifying Players and Playfield: To classify the players and the playfield, we follow the well-known object detection framework proposed by Viola and Jones [6] and its succeeding developments proposed by many other researchers [7]–[9] to design our player detector. Due to the speciality of our problem, we make several improvements to make the player detectors much more efficient and effective.

The first improvement lies in the weak feature level. Previous studies on object detection prove that color information (in gray level) is effective for face detection [6], [7] while gradient information plays an important role in pedestrian detection [8], [9]. Although the players in sports videos are much more complex than faces and pedestrians, we can still make an analogy from the players to faces and pedestrians. On the one hand, color information can be useful for player detection as in face detection since the color of the players shares a similar distribution; on the other hand, players exhibit different poses like pedestrians, therefore gradient information can be effective to capture the contour shape. Thus, using both color and shape information to train the player detector can be more effective. We therefore extend the feature in [7] by adding gradient information to the weak feature to train the detector.

The second improvement comes from the bootstrap procedure in the boosting framework. In most previous works of object detection, the negative samples in the boosting training are selected from a large set of the background image pool, which is collected from different kinds of images without positive samples. In our problem, since the player detector only needs to distinguish between players and the playfield, the negative samples are only selected from images with an empty playfield (no player appears). Because the playfields share a rather similar and simple distribution, the training of the player detector can

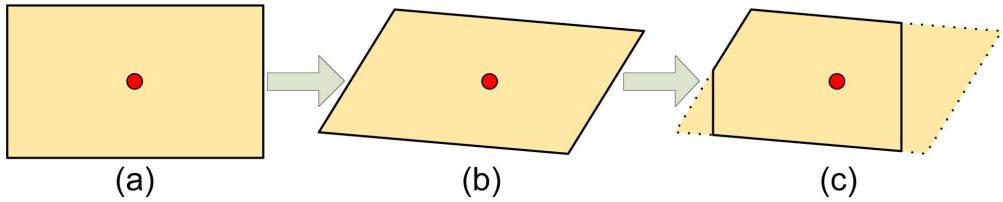

Fig. 4. Shape of the playfield viewing from a camera. (a) Actual playfield shape. (b) Playfield shape through a camera. (c) Visible playfield region in one frame.

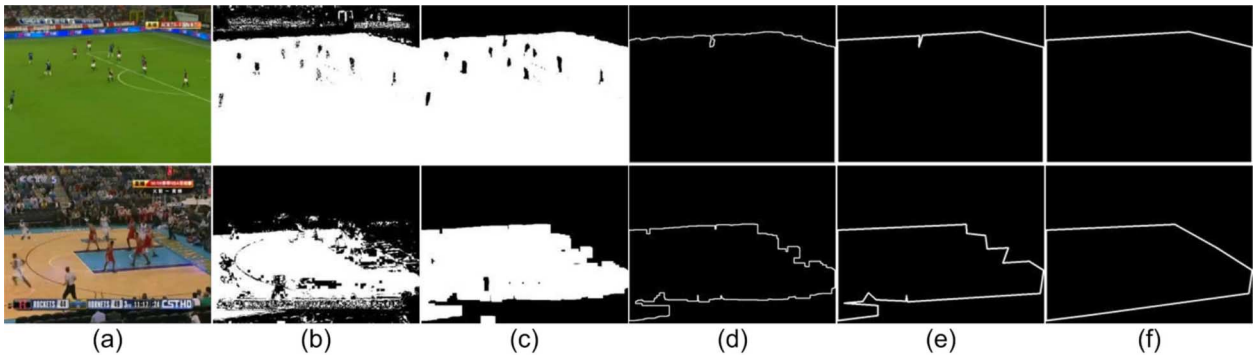

Fig. 5. Illustration of the playfield contour extraction process. (a) Input frame. (b) After ST-GMM segmentor. (c) After temporal propagation. (d) Extracted maximal contour. (e) After edge compact. (f) Convex hull as final playfield contour.

converge much faster, and the obtained detector is much more efficient in online classification.

The last improvement is the detector structure and object samples. Based on the player characteristic in different types of sports video, a corresponding detector structure and object sample is selected. Specifically, for the football and hockey players that usually exhibit full body in a distant view, the cascade structure in [6] is adopted, and the full body of the player is cropped as a training sample; for basketball players that are usually shown with different poses in a close shot, the vector tree structure with three different views in [7] is employed, and the upper player body is selected as the object sample. These treatments are not trivial because we find that they have significant impact on the detector performance through the experiments.

Based on these improvements, the trained player detector obtains a great performance improvement and functions effectively and efficiently in the whole system.

3) Classifying Player and Other Players: The last step of the progressive observation modeling process is to classify one player from other players. To distinguish between different players, the basic idea is to learn a discriminative observation model for each player by exploring different kinds of object-specific information and continuously refine it through online updating. Based on this thought, we build a dedicated observation model for each player when a new object trajectory is initialized and update it during the trajectory growing process. The dedicated observation model maintains a set of discriminative cues to model the player's special feature. In particular, three different types of information, including appearance, size, and motion, are explored. The player dedicated model can be represented as

$$P_k = (\mathbf{a}_k, x_k, y_k, w_k, h_k, v_{k,x}, v_{k,y}) \tag{9}$$

TABLE I
SUMMARY OF THE OBSERVATION MODELING PROCESS

Step	Description
1	Classifying playfield and non-playfield: Offline ST-GMMs segmentor + online motion and shape model.
2	Classifying players and playfield region: Offline boosted player detectors.
3	Classifying one player and other players: Online learned discriminative appearance, size, motion model.

where $\mathbf{a}_k$ is the object appearance, x_k, y_k, w_k, and h_k are the player x location, y location, width, and height, which are provided by the player detector, and $v_{k,x}$ and $v_{k,y}$ are the velocities that are calculated from the target moving trajectory and updated by the Kalman filter [12]. The object appearance is modeled as the color histogram in the discriminative region of the object. One way to learn the discriminative region is to online select the discriminative features between different players as proposed by the methods in [35] and [10]. Another way is to cooperate the domain-dependent knowledge to make the learning process more simple and efficient. For sports video, the clothes region of a player is the most distinctive region in distinguishing this player from players of another team. Thus, we locate the clothes region of the player based on the player detector and calculate its color histogram in HSV color space learned when it is isolated from other targets. In implementation, the number of bins for each color channel is set to 16. Since the HSV color space decouples the intensity (i.e., value) from color (i.e., hue and saturation), the histogram gets $272(= 16 \times 16 + 16)$ bins in total.

In Table I, we give a summative description of the above three steps.

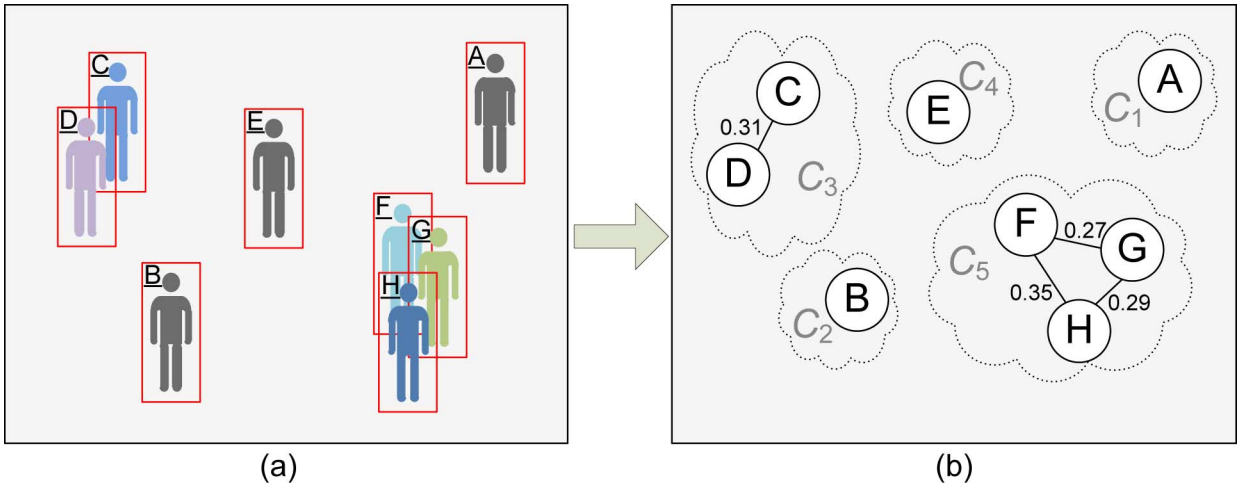

Fig. 6. Modeling the object occlusion relationship using an undirected graph. By finding the connected components of the graph, each object is either in an isolated cluster or in an occluded cluster. (a) Object detection and tracking results. (b) Undirected occlusion relationship graph.

B. Dual-Mode Two-Way Bayesian Inference

Based on the observation modeling process, we propose to track a single isolated player and multiple occluded players in a dual-mode integral tracking framework to deal with the observation assignment problem. Specifically, it has two different tracking modes: the independent tracking mode (ITM) and the simultaneous tracking mode (STM). In the ITM, each isolated player is tracked through a general player model independently, while, in the STM, multiple occluded players in one occlusion cluster are tracked simultaneously through the dedicated player models. What is more, we also introduce a backward smoothing procedure to handle the observation missing problem.

1) Dual-Mode Integral Tracking: In order to track a player with its corresponding mode, we build an undirected graph to model the occlusion relationships between different players. Supposing there are M players at one frame, the occlusion graph is represented as $G = \{r_{ij}\}_{i,j=1}^{M}$, where r_{ij} is the graph node reflecting the relationship between the ith player and the jth player. The graph is first initialized by calculating the overlap ratio between two players and then updated during the integral tracking process. By traversing the graph to find its connected components, each node is classified to be either an isolated node or an occluded one. For each isolated node, the tracking mode is set to be the ITM, and for all occluded nodes in each occlusion cluster, the tracking mode is set to be the STM. Fig. 6 gives an illustration of the occlusion relationship graph.

In the ITM, the observation model is generated from the offline learned player detector which returns a confidence value reflecting its likelihood of a particle. Considering the characteristic of a boosted object detector, i.e., the more layers of classifiers a particle passed and the larger the marginal distance when it passed, the more likely that the particle contains the object, the observation model in the ITM is represented as

$$L^{\mathrm{ITM}}(\mathbf{o}_t|\mathbf{s}_t) = \frac{\eta \exp(U)}{\exp(L_T - L_P)} \tag{10}$$

where L_T is the total layer number of the detector, L_P is the layer number that the observation passed, U is the sum of the marginal distance when passing one layer, and η is a normalization factor which makes the likelihood to be a probability distribution.

In the STM, all of the players in one occlusion cluster are tracked simultaneously. The particles of these players are evaluated by a candidate observation model which is generated from the online learned player model for each player. We denote a particle as $(x_n, y_n, w_n, h_n, \pi_n, l_n)$, where x_n, y_n, w_n, and h_n indicate its location and size, π_n is the particle weight, and l_n is a label reflects which player it belongs to. Taking one occlusion cluster C_z and its corresponding particle set $\{\mathbf{s}_{C_z}^n, \pi_{C_z}^n\}$ (subscript t is suppressed here for clarity), for example, the affinity between a particle $\mathbf{s}_{C_z}^n$ and a player P_k in cluster C_z is modeled as

$$\pi_{n,k} = \alpha \mathbf{A}\left(\mathbf{s}_{C_z}^n, P_k\right) + \beta \mathbf{S}\left(\mathbf{s}_{C_z}^n, P_k\right) + \gamma \mathbf{M}\left(\mathbf{s}_{C_z}^n, P_k\right) \tag{11}$$

where the weighting terms $\mathbf{A}(\cdot)$, $\mathbf{S}(\cdot)$ and $\mathbf{M}(\cdot)$ model the affinity between an observed particle and a player in appearance, size, and motion respectively, and the weighting factor α, β, and γ decide the relative importance of each model. In implementation, $\mathbf{A}(\cdot)$ is measured by the appearance similarity using the Bhattacharyya coefficient between the color histograms of the image region within the particle and the object color model. Denoting Δt as the frame gap between the particle sample time and the player model time, $\mathbf{S}(\cdot)$ and $\mathbf{M}(\cdot)$ is modeled as

$$\mathbf{S}\left(\mathbf{s}_{C_z}^n, P_k\right) = \exp\left(-\frac{|w_n - w_k| + |h_n - h_k|}{w_n + w_k + h_n + h_k}\right) \tag{12}$$

$$\mathbf{M}\left(\mathbf{s}_{C_z}^n, P_k\right) = \exp\left(-\frac{|x_n - x_k| + |y_n - y_k|}{|v_{k,x}\Delta t| + |v_{k,y}\Delta t|}\right). \tag{13}$$

Thus, the observation model of P_k in the STM can also be represented as

$$L^{\mathrm{STM}}(\mathbf{o}_t|\mathbf{s}_t) = \alpha \mathbf{A}(\mathbf{o}_t, P_k) + \beta \mathbf{S}(\mathbf{o}_t, P_k) + \gamma \mathbf{M}(\mathbf{o}_t, P_k). \tag{14}$$

By calculating the affinities with all of the players in the cluster, the particle weight in STM stage is set to the largest affinity

$$\pi_n^{\mathrm{STM}} = \max\{\pi_{n,k}\}_{k \in C_z} \tag{15}$$

and the label of the particle is decided by

$$l_n = \arg\max_k \{\pi_{n,k}\}_{k \in C_z}. \tag{16}$$

Then, each player in the cluster is tracked using its supporting particles.

2) Two-Way Bayesian Inference: When one player undergoes abrupt changes or is serious occluded by other objects, the dual-mode integral tracking process may fail in tracking it because no observation of this player could be collected. This is what we named as the observation missing problem which often causes a track fragmentation that many MOT algorithms suffer.

To surmount this problem, a backward smoothing process is adopted to retrack the lost player. During the integral tracking, when a player cannot get enough supporting particles (the sum of supporting particle weights is below a threshold T_S or the number is below a threshold T_N), the track of the player is buffered for future backward smoothing with newly collected observations. Denoting the state of buffered player set as $\mathbf{S}^B = \left\{\mathbf{s}_i^B\right\}_{i=1}^{M_B}$ with its dedicated model set $\left\{P_i^B\right\}_{i=1}^{M_B}$ and the new object observation set $\mathbf{O}^N = \left\{\mathbf{o}_j^N\right\}_{j=1}^{M_N}$ with its dedicated model set $\left\{P_j^N\right\}_{j=1}^{M_N}$, the backward smoothing procedure is performed on these two sets. It first finds the match between them based on their affinity and then smooths backward from the new object to the buffered object. The affinity model between P_i^B and P_j^N is similar to the observation model in the STM, which can be represented as

$$A_{i,j} = \alpha \mathbf{A}\left(P_i^B, P_j^N\right) + \beta \mathbf{S}\left(P_i^B, P_j^N\right) + \gamma \mathbf{M}\left(P_i^B, P_j^N\right). \tag{17}$$

Based on this affinity model, the matching score matrix can be constructed as $C = \{A_{i,j}\}$, and the optimal match between these two sets can be obtained through the Hungarian algorithm [36]. However, this "optimal" match tends to overmatch the new object observation set to the buffered object set which corresponds to the situation that a new object observation and a buffered object is matched with a relatively low affinity. This is because the Hungarian algorithm is designed to maximize the total matching score and there is no mechanism to impose a penalty on the overmatch. To alleviate this problem, we introduce a threshold function to the matching score matrix and make the overmatch avoidable in the matching result. Denoting the new matching score matrix as $C' = \{T(A_{i,j})\}$, $T(\cdot)$ is the threshold function with the following form:

$$T(x) = \begin{cases} x, & \text{for} x \geq T_M \\ 0, & \text{for} x < T_M \end{cases} \tag{18}$$

where T_M is the matching affinity threshold which prevents a match with an affinity below it. Based on the matching result of applying Hungarian algorithm on C', the object trajectory initialization, growth, and termination are managed integrally as follows.

1) If $\mathbf{s}_{t,y}^B$ and $\mathbf{o}_{t,x}^N$ are matched, grow the trajectory of $\mathbf{s}_{t,y}^B$ with $\mathbf{o}_{t,x}^N$.
2) If $\mathbf{o}_{t,x}^N$ cannot get matched, initialize a new trajectory.
3) If $\mathbf{s}_{t,y}^B$ cannot get matched in T_W frames, terminate its trajectory.

For easy reference, we summarize the entire procedure of the proposed MOT algorithm in Table II.

V. Experiments

Here, implementation details of the system and experimental results will be presented. We first give a short description of the sports video data used in the experiments. Then, we illustrate the training process of the offline models. Following is the evaluation of the system performance. Finally, some discussions of the proposed system are presented.

A. Sports Video Data

One of the motivation of this work is to track multiple players in videos of football and basketball, which are two typical sports and have gained much popularity around the world. Thus, the experiment results are mainly demonstrated on these two kinds of sports video. The football and basketball video data used for training and testing are both directly collected from the recorded videos of the sports league match, such as the European Champion's League and the National Basketball Association, which represent the highest level of the game. For each sport, we collected ten sequences respectively, among which five sequences are used for training and the other five are used for testing. We manually label the ground truth of both the training and test sequences for quantitative analysis. Besides football and basketball, for a fair comparison of our algorithm with the representative work in [25], which deals with the problem of multiple hockey players tracking, we will also give the result of our algorithm on hockey videos. The training and testing video data for hockey player tracking are kindly provided by the authors in [25].

B. Offline Observation Models Training

There are two offline models in the progressive observation modeling process: the playfield classification model and the player detector model.

The playfield classification model is learned from some sports video frames that expose different shapes of the playfield. For each frame, we first simply label the vertices of the playfield and then generate the polygon mask of the playfield region. In total, 500 frames were labeled for football games and basketball games, respectively. Fig. 7 shows some typical training samples. The ST-GMM model was trained for each location using the EM algorithm with sampling in the spatial temporal space and dynamically determining the number of GMM components as described in Section IV-A1.

The player detector model is trained from a number of labeled positive and negative player samples in a fixed size. Considering the camera settings, the training sample size for football player detector is set as 20×30 pixels which contains the full player body and for basketball player detector it is set as 24×24 pixels which only contains the head–shoulder region. Three thousand football player samples and 5000 basketball player samples are labeled to train corresponding detectors. Negative training samples are collected from a large background image pool which contains only the playfield region without any players. Fig. 8 shows some typical player

TABLE II
PROPOSED TRACKING ALGORITHM

Input: frame data I, algorithm parameters (weighting factors α β γ in STM particle evaluating, buffer waiting size T_W, particle number N for each object, supporting particle weight sum threshold T_S, supporting particle number threshold T_N, backward smoothing matching threshold T_M).

Given: the tracked object set $\mathbf{S}_{t-1} = \{\mathbf{s}_{t-1,m}\}_{m=1}^{M}$ with the supporting particle set $\{\mathbf{s}_{t-1,m}^{n}, \pi_{t-1,m}^{n}\}_{n=1}^{N}$ and the dedicated player model $\{P_m\}_{m=1}^{M}$ of each object, buffered object set $\mathbf{S}_{t-1}^{B} = \{\mathbf{s}_{t-1,m}^{B}\}_{m=1}^{M_B}$, occlusion graph G and the playfield region at frame $t-1$, proceed at frame t as follows:

- For each tracked object $\mathbf{s}_{t-1,m}$:
  - + If $\mathbf{s}_{t-1,m}$ is an isolated object, set its tracking mode to ITM and track it using the general observation model:
    - - Resample: simulate $a_n \sim \{\mathbf{s}_{t-1,m}^{n}, \pi_{t-1,m}^{n}\}_{n=1}^{N}$, and replace $\{\mathbf{s}_{t-1,m}^{n}, \pi_{t-1,m}^{n}\}_{n=1}^{N}$ with $\{\mathbf{s}_{t-1,m}^{a_n}, \pi_{t-1,m}^{a_n}\}_{n=1}^{N}$;
    - - Prediction: simulate $\mathbf{s}_{t,m}^{n} \sim D(\mathbf{s}_{t,m}|\mathbf{s}_{t-1,m}^{n})$;
    - - Update: $\pi_{t,m}^{n} \leftarrow L^{\mathrm{ITM}}(\mathbf{o}_{t,m}|\mathbf{s}_{t,m}^{n})$;
    - - If $\sum_{n=1}^{N} \pi_{t,m}^{n} > T_S$, object $\mathbf{s}_{t,m}$ is tracked, else buffer it;
    - - Inference: $\hat{\mathbf{s}}_{t,m} \leftarrow \sum_{n=1}^{N} \pi_{t,m}^{n} \cdot \mathbf{s}_{t,m}^{n}$;
    - - Update the candidate model of object $\mathbf{s}_{t,m}$;
  - + Else $\mathbf{s}_{t-1,m}$ is an occluded object in cluster C_z, set its tracking mode to STM and track them simultaneously using the dedicated models:
    - - Resample: simulate $a_n \sim \{\mathbf{s}_{t-1,C_z}^{n}, \pi_{t-1,C_z}^{n}\}_{n\in C_z}$, and replace $\{\mathbf{s}_{t-1,C_z}^{n}, \pi_{t-1,C_z}^{n}\}_{n\in C_z}$ with $\{\mathbf{s}_{t-1,C_z}^{a_n}, \pi_{t-1,C_z}^{a_n}\}_{n\in C_z}$;
    - - Prediction: simulate $\mathbf{s}_{t,C_z}^{n} \sim D(\mathbf{s}_{t,C_z}|\mathbf{s}_{t-1,C_z}^{n})$;
    - - Update: $\pi_{t,C_z}^{n} \leftarrow L^{\mathrm{STM}}(\mathbf{o}_{t,C_z}|\mathbf{s}_{t,C_z}^{n})$;
    - - If $\sum \delta_m(l_n)\pi_{t,C_z}^{n} > T_S$ and $\sum \delta_m(l_n) > T_N$, object $\mathbf{s}_{t,m}$ is tracked, else buffer it;
    - - Inference: $\hat{\mathbf{s}}_{t,m} \leftarrow \sum \delta_m(l_n) \cdot \pi_{t,C_z}^{n} \cdot \mathbf{s}_{t,C_z}^{n}$;
- Segment playfield using the offline classification model to extract the playfield contour;
- Search for the new object set $\mathbf{O}_t^N = \{\mathbf{o}_{t,m}^N\}_{m=1}^{M_N}$ within the playfield region;
- Backward smoothing between $\mathbf{S}_{t-1}^B$ and $\mathbf{O}_t^N$;
  - + If $\mathbf{s}_{t-1,i}^B$ and $\mathbf{o}_{t,j}^N$ are matched, link them and add to $\mathbf{S}_t$;
  - + Else If $\mathbf{s}_{t-1,i}^B$ can't get matched, terminate it if its waiting time exceeds T_W;
  - + Else If $\mathbf{o}_{t,j}^N$ can't get matched, initialize a new track using $\mathbf{o}_{t,j}^N$ and add it to $\mathbf{S}_t$;
- Updating the occlusion graph G based on $\mathbf{S}_t$;

Output: the playfield region with the tracked players in it.

training samples. To evaluate the detection performance of the trained player detectors, We build evaluation data sets for both football players and basketball players. For each type of sports, the evaluation set contains 100 images with ground truth of more than 500 players in different views and poses. To make a comparison with the state-of-the-art detector, we trained another boosted object detector using the HOG feature proposed in [9] which has been proven to be very effective for human detection. Fig. 9(a) shows the ROC curves between detection rate and number of false alarms and Fig. 9(b) gives the ROC curves between detection rate and false alarm rate. From Fig. 9, we can see that our detectors have a relatively higher detection rate than the HOG boosted detector at the same false alarm level. We attribute this to the progressive observation modeling process, since it makes the player detector only needs to distinguish the players and playfield region.

C. Performance Evaluation

We first conduct experiment to analyze the STM sampling process of our algorithm. The experiment is carried on tracking two football players who approach each other and then split away. The whole procedure takes about 40 frames which is very typical in sports videos. Fig. 10(a) gives a vivid example of how

Fig. 7. Training samples of the playfield segmentor. First line: typical frames collected from sports videos; second line: labeled playfield regions (in white).

Fig. 8. Training samples of the player detectors. The first and third lines are the football and basketball player samples, respectively; the second and fourth lines are the corresponding negative samples collected in the training process.

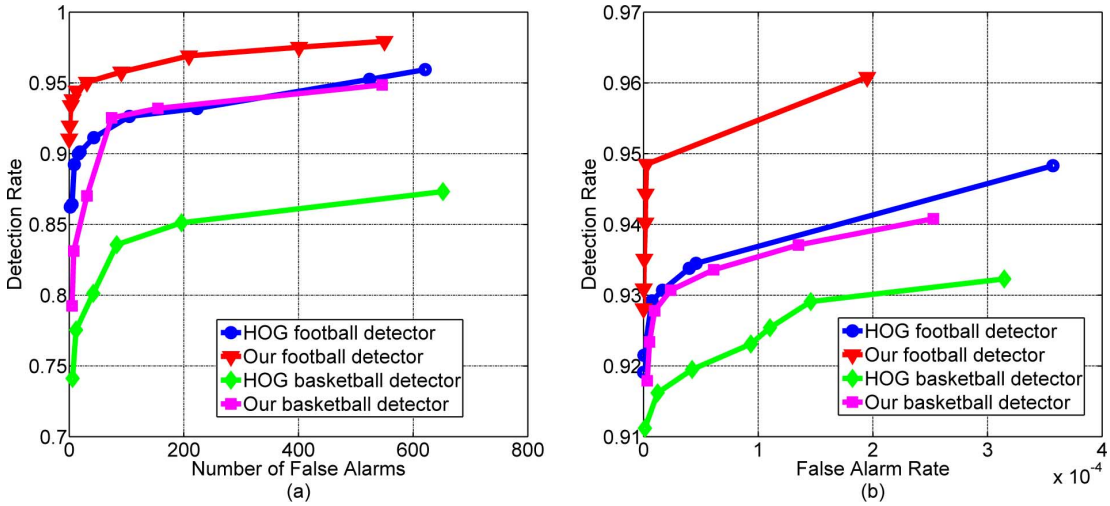

Fig. 9. Evaluation of the player detectors. (a) ROC curve between Detection Rate and Number of False Alarms tested in a 100 evaluation set. (b) ROC curve between Detection Rate and False Alarm Rate tested in the positive and negative sample sets.

our STM sampling process works for tracking these two occluded players in the 18th frame with comparison to the standard particle filter (SPF) [9] which tracks them independently with a general player detector. From Fig. 10(a), we can see that our STM sampling process can well distinguish the two objects and avoid the drifting tendency in the SPF sampling process. In Fig. 10(b), we give the tracking accuracy in object position and size by comparing the tracking results with labeled ground truth which proves that our algorithm can give a more accurate and tighter object bounding box especially when occlusion happens.

We further conduct experiments to compare our work with the standard particle filter algorithm [9] and the two-stage MOT algorithm proposed in [11]. We implement the standard particle filter algorithm for MOT by initializing an independent tracker for each object. For the two-stage tracking algorithm (TS), which uses part detectors to collect observation and does

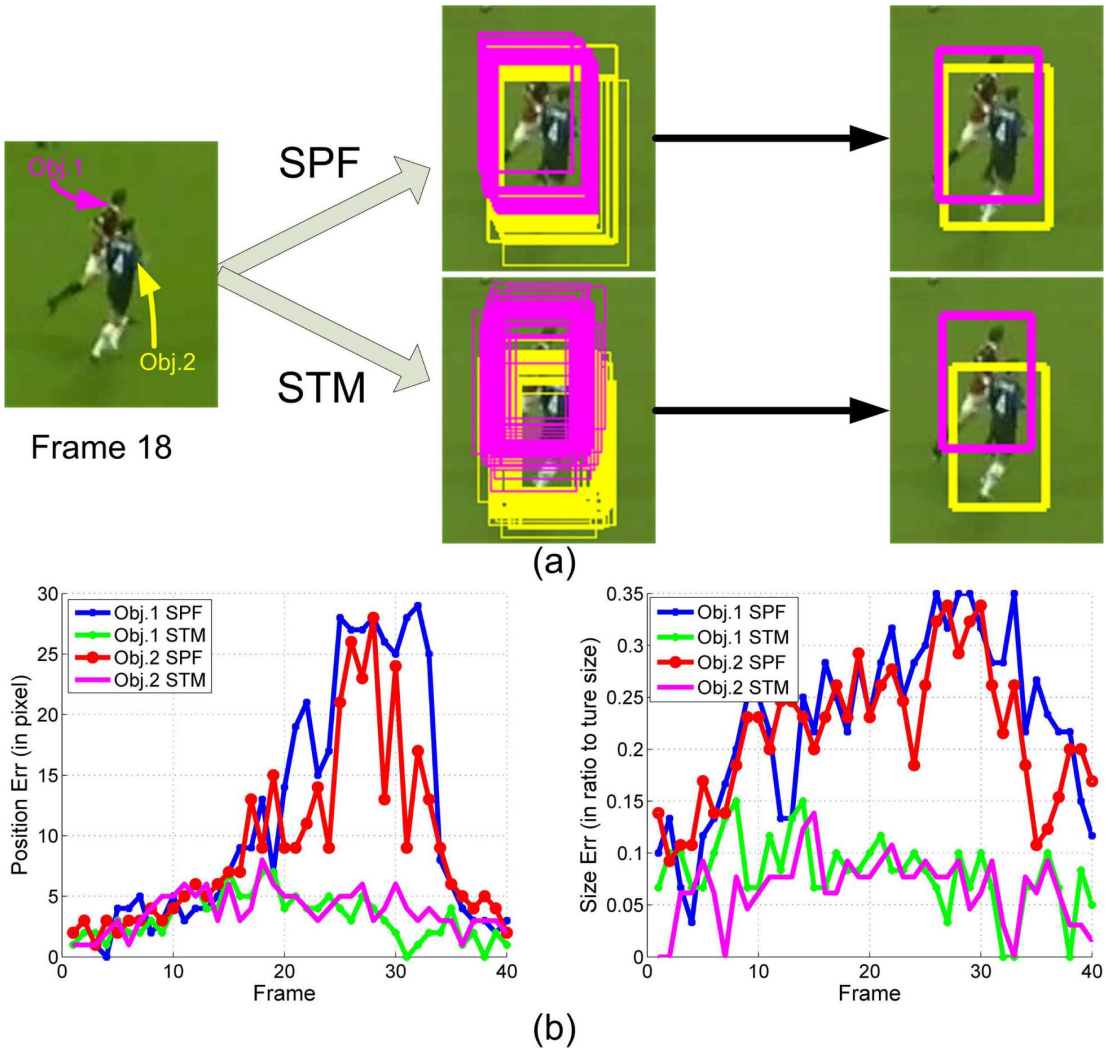

Fig. 10. Comparison between the STM sampling process and the SPF sampling process when tracking two occluded players. (a) Comparison of sampling process. (b) Comparison of tracking accuracy for one player in 40 consecutive frames.

association in a temporal sliding window to deal with temporary occlusion, we implement it with the suggested settings as in [11]. For fair comparison, all the player detectors in the three algorithms are trained using the same method with the same feature and the playfield segmentation result of our algorithm is provided to them to remove detection responses outside the playfield. We adopt the same metrics for evaluating tracking performance as in [8] and [11] since they provide a straight-forward and effective measurement both in the observation model level and the tracking strategy level. These metrics are defined as follows:

- **MT**: the number of Mostly Tracked trajectories (trajectories that are tracked for more than 80% of the frames).
- **PT**: the number of Partially Tracked trajectories (trajectories that are tracked between 20% and 80% of the frames).
- **ML**: the number of Mostly Lost trajectories (trajectory that is lost more than 80% of the frames).
- **IDS**: the number of ID Switches (two trajectories exchange their IDs).
- **FRMT**: the number of Fragmentations (trajectory that is broken off).

TABLE III
EVALUATION RESULTS

Sequences	Algorithm	MT	PT	ML	IDS	FRMT
Football	SPF[13]	91	29	14	42	48
	TS[11]	101	26	7	18	27
	Proposed	119	12	3	12	18
Basketball	SPF[13]	21	23	17	47	51
	TS[11]	38	13	10	27	34
	Proposed	45	12	4	20	25

The test videos are five football sequences and five basketball sequences with labeled ground truth. The five football sequences contain 134 object trajectories and last for 2593 frames, and the five basketball sequences contain 61 object trajectories and last for 1572 frames. The evaluation results are given in Table III, from which we can see that the proposed tracking algorithm makes a significant improvement on all metrics compared with the standard particle filter algorithm and the two-stage tracking algorithm in [11]. The improvements

Fig. 11. Example tracking results. The first and third lines are the tracking results using the standard particle filter algorithm [13] and the two-stage algorithm [11], respectively. The second and fourth lines are the tracking results using the proposed algorithm.

on the metric MT, PT, and ML are mainly attributed to the mechanism of the progressive observation modeling, and the reduce of IDS and FRMT number proves the effectiveness of the proposed Bayesian inference framework to the observation assignment and missing problem. In Fig. 11, we give some example tracking results. It can be seen that our algorithm works robustly under many very difficult situations, such as abrupt pose changes of the player (e.g., Player 6 in Frame 070 and Player 5 in Frame 083), complex objects interactions (e.g., Players 8 and 9 in Frame 040, Players 3 and 6 in Frame 060, and Players 3 and 4 in Frame 098).

At last, we conduct experiments on hockey player tracking and give a quantitative comparison with the method in [25]. The positive and negative training samples of the player detectors and the test sequences are kindly provided by the authors of [25]. We use these samples to train the player detector model and use the negative samples to train the playfield classification model as in the football and basketball sequences. We manually label the tracking ground truth of the test sequences and evaluate the performance of our algorithm both on the observation level and tracking level for each frame. We compare our result with that presented in the supplementary material[1] of [25]. The three evaluation metrics used are the True Detection Number (DN), False Alarm Number (FN) and Tracking Failure Number (TN). The DR and FN are defined just as their names which reflect the observation performance. The TN is to evaluate the tracking performance which is defined as the number of tracking failures,

[1][Online]. Available: http://www.cs.ubc.ca/ōkumak/research.html

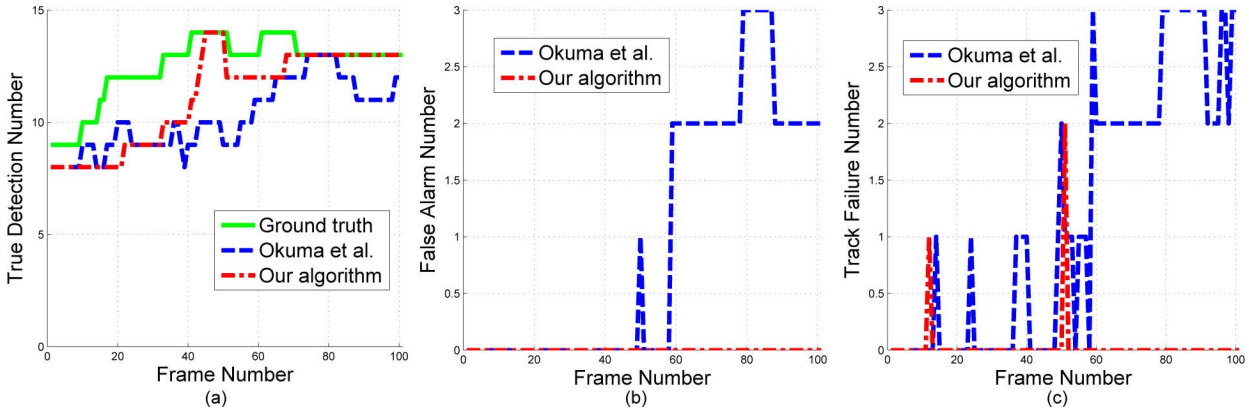

Fig. 12. Performance comparison between Okuma *et al.* [25] and our algorithm. (a) True Detection Number at each frame. (b) False Alarm Number at each frame. (c) Track Failure Number at each frame.

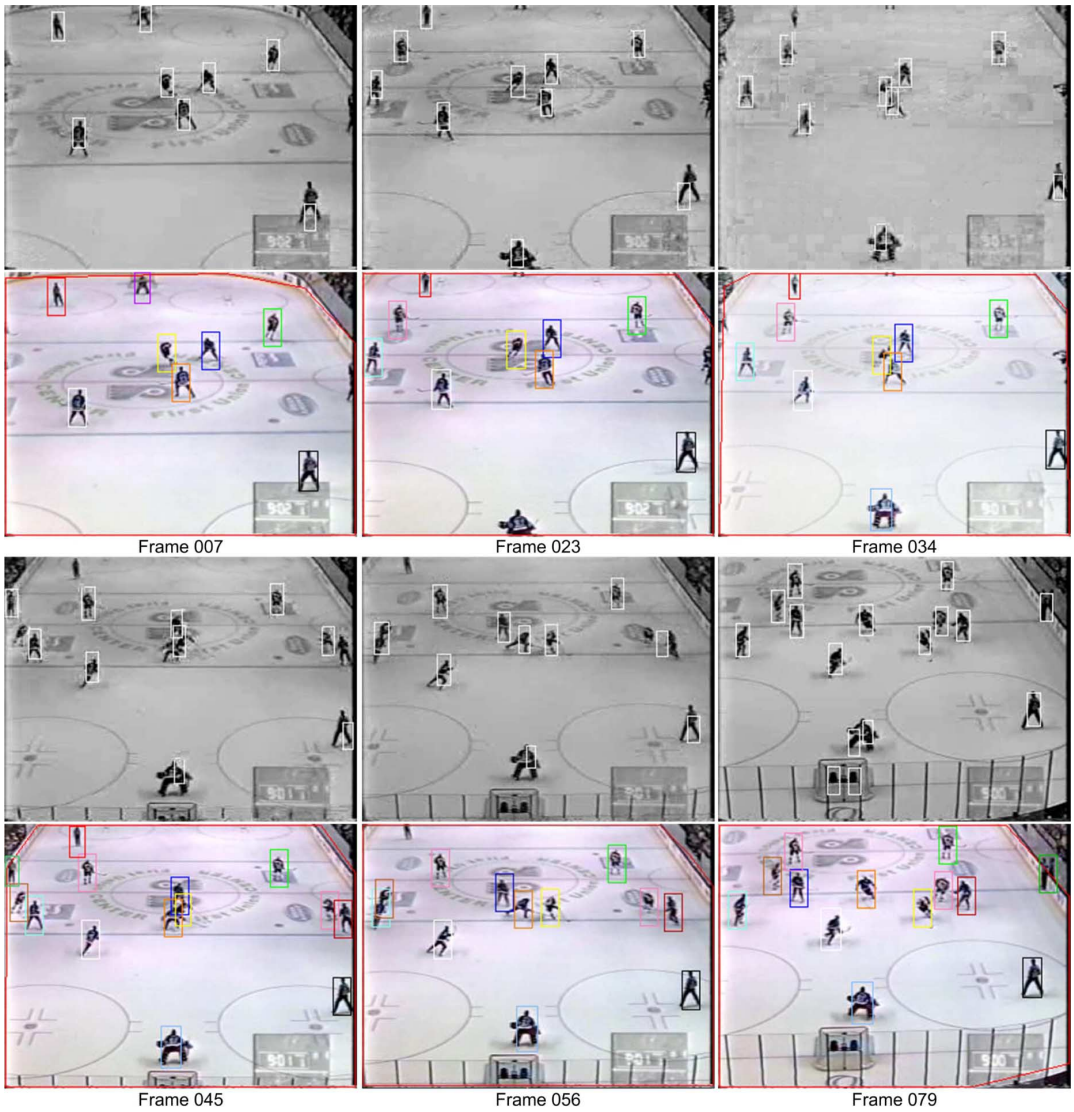

Fig. 13. Comparison experiment on hockey videos. The first and third lines are the result given by the algorithm in [11]. The second and fourth lines show the result obtained by the proposed algorithm.

including lost tracks, false tracks and ID switches. In Fig. 12, we draw the value of these three metrics at each frame. As shown in Fig. 12(a) and (b), our algorithm has obtained a relatively higher detection rate than the method in [25] while suppressing all the false alarms. It further proves the effectiveness of the progressive observation modeling mechanism. From Fig. 12(c),

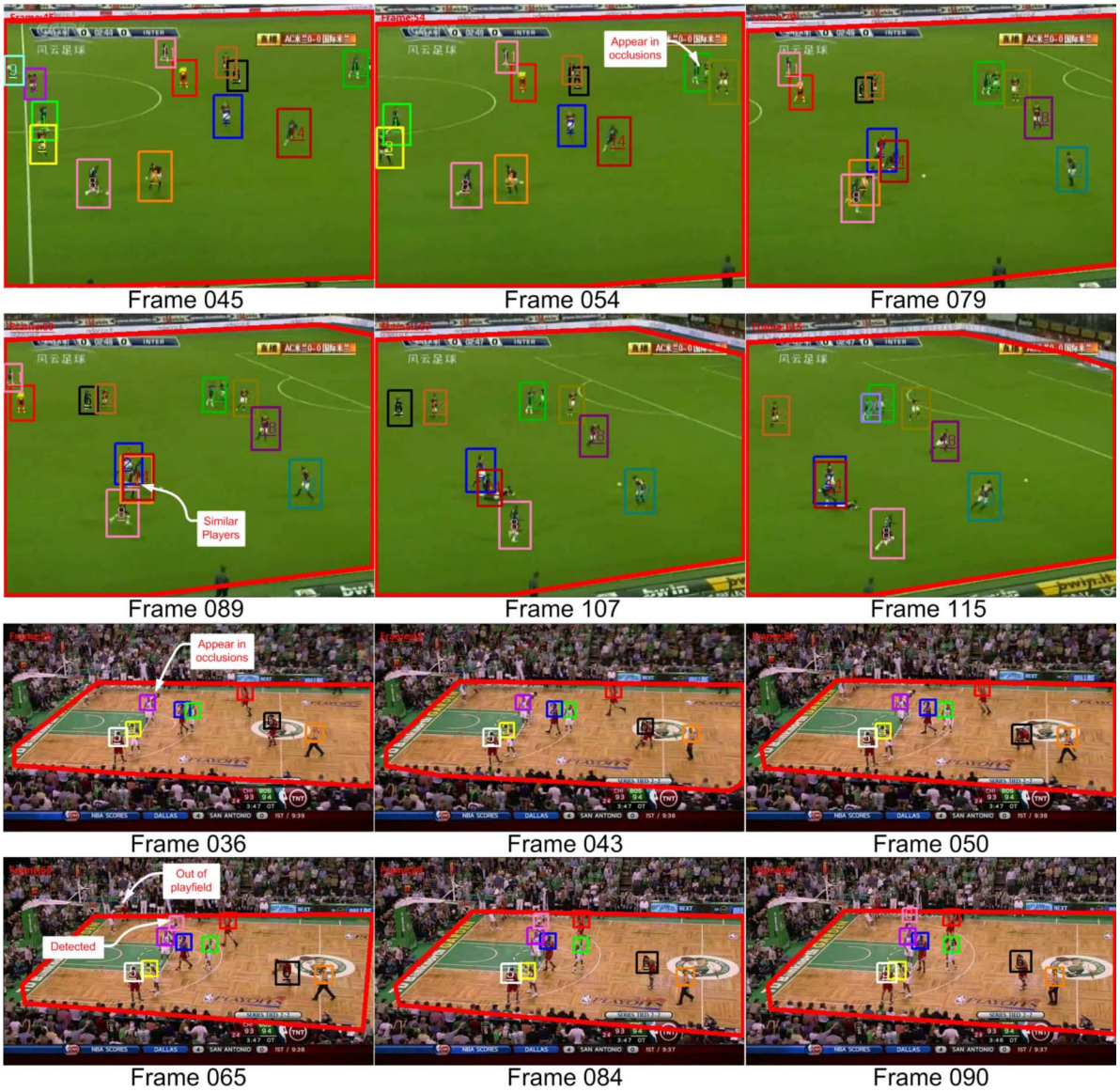

Fig. 14. Tracking results of the proposed algorithm on another football and basketball video sequences.

we can see that our algorithm is also superior to [25] on the tracking level. Our algorithm can overcome most of the difficult situations and make reliable tracking. Some visual examples of this experiment are given in Fig. 13.

D. Discussions

Although all of the above experiments have validated the proposed approach, some other issues need to be discussed furthermore.

Algorithm Parameters: In the proposed tracking algorithm, the weighting factors α, β, and γ reflect the relative importance of the color, size, and motion cue in the affinity model. In our experiments, we set it as $\alpha : \beta : \gamma = 5 : 1 : 3$ for the football and hockey videos and $\alpha : \beta : \gamma = 5 : 1 : 2$ for the basketball videos. The value of the particle number N is set to 200, and the supporting particle threshold T_S and T_N are set as $0.4N$ and $0.6N$ proportionately. The buffer waiting size T_W is set as 100. For the matching threshold T_M, the value 0.8 could avoid most of the false matches in our experiments. In training the offline ST-GMM segmentation model, the merge and split threshold T_G and T_P are set as 15 and 60, respectively. The learning rate λ for the motion model is set as 0.2.

1) Processing Speed: The current system is implemented in C++ without special code optimization. The experiments run fluently on most of the test videos using a 2.40-GHz. Intel Core 2 CPU. Three main factors that affect the processing speed are the offline player detector complexity, the resolution of the video, and the player number. Since the basketball player detector is the most difficult to train, the average processing speed on basketball videos with resolution 616×472 is approximately 10 fps. For the football videos with the same resolution, the processing speed is about 15 fps. For the hockey videos, whose resolution is 320×240, the system runs at about 30 fps.

2) Failure Cases: The observation modeling process greatly improves the discriminative power of the system measurements. In some cases, however, it may also fail to observe the target.

When two targets come along together with serious occlusion, the observation model is likely to miss one of them (Football Frame 054 and Basketball Frame 036 in Fig. 14). Another situation that the observation model may fail is when the target appears outside of the playfield region (Basketball Frame 65 in Fig. 14). The observation model will treat it as a false alarm due to the restriction from the playfield contour. For all of these cases, the system has to wait for the target to come out of occlusion or move into the playfield region (Basketball Frame 065 in Fig. 14). In the tracking level, current implementation uses the affinities in appearance, size and motion to distinguish different targets. When two targets are similar in these aspects, the tracking procedure could fail to distinguish them or make false backward smoothing (One case for this kind of failure can be seen at Football Frame 089 in Fig. 14). To remedy this, employing more discriminative features and more sophisticated feature selection schema will be helpful.

VI. Conclusion

In this paper, we focus on the challenging problem of tracking multiple highly dynamic and highly interactive players in sports video. Facing the difficulties it presents, we propose a new algorithm that formulates the SOT problem and MOT problem in a unified dual-mode two-way Bayesian inference approach based on the observations that are built progressively. The proposed method obtains satisfactory tracking results on many typical real world sports videos. Although the current system is demonstrated on sports videos, we believe that the proposed multiple object tracking strategy and the progressive observation modeling mechanism can be easily extended to other dynamic and interactive scenes. Our future work will be carried out on applying the proposed method to other scenes to further test its adaptability. Some high-level analysis of the object behaviors based on the tracking results can also be investigated.

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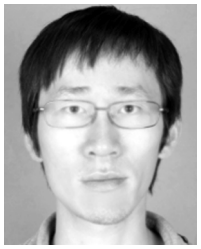

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